Case 1: $r_1 - r_2 = 2\nu$ Is Not a Non-Negative Integer

- To find a second, independent solution, we must consider **three cases** for the indicial-root difference $r_1 - r_2 = 2\nu$.
- A second Frobenius series solution for $r_2 = -\nu$ exists. Repeating the identical calculation with $\nu \to -\nu$ throughout gives the second series solution

$$\boxed{J_{-\nu}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n-\nu+1)} \left(\frac{x}{2}\right)^{2n-\nu}} \qquad (4)$$

- Note that $J_{-\nu}(x)$ is unbounded at $x = 0$ if $\nu > 0$.
- Thus $J_\nu(x)$ and $J_{-\nu}(x)$ are linearly independent, so the general solution of (1) is

$$y(x) = C_1 J_\nu(x) + C_2 J_{-\nu}(x).$$



Case 2 and Case 3: Setup

- **Case 2:** $r_1 = r_2$**.** This corresponds to $\nu = 0$, and a second Frobenius series solution does **not** exist directly from this method.
- **Case 3:** $r_1 - r_2 = 2\nu$ **is a positive integer.** There are two subcases.
- **Subcase (a):** ν **is not a positive integer.** Then clearly $\nu = \dfrac{2k+1}{2}$ for some $k \in \{0, 1, 2, \dots\}$ (i.e. ν is a positive half-integer).
- As before (since $a_0 \neq 0$),

$$\rho(r) = 0, \qquad \rho(r+1)a_1 = 0, \qquad \rho(r+n)a_n = -a_{n-2}, \quad n \geq 2.$$



Subcase (a): Half-Integer ν — Second Solution Still Exists

- With $r = r_2 = -\nu$, we get

$$\rho(r) = 0; \quad 1\cdot\big(1-(2k+1)\big)a_1 = 0; \quad n\cdot\big(n-(2k+1)\big)a_n = -a_{n-2},\ n \geq 2.$$

- All even terms a_{2n} are determined uniquely. For odd terms, $a_1 = a_3 = \cdots = a_{2k-1} = 0$, but at a_{2k+1} we must have

$$n \cdot 0 \cdot a_{2k+1} = -a_{2k-1} \implies 0 \cdot a_{2k+1} = 0.$$

- This holds for *any* value of a_{2k+1}; for simplicity take $a_{2k+1} = 0$, so the rest of the odd terms vanish too.
- Thus the second solution in this case is again given by (4):

$$J_{-\nu}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n-\nu+1)} \left(\frac{x}{2}\right)^{2n-\nu}. \tag{5}$$



Subcase (b): ν a Positive Integer — No Second Frobenius Solution

- **Subcase (b): ν is a positive integer.** Clearly $\nu = k$, $k \in \{1, 2, 3, \dots\}$.
- With $r = r_2 = -\nu$, we get

$$\rho(r) = 0; \quad 1 \cdot (1 - 2k)a_1 = 0; \quad n \cdot (n - 2k)a_n = -a_{n-2}, \quad n \geq 2.$$

- All odd terms a_{2n+1} vanish. Since $a_0 \neq 0$, the even terms $a_2, a_4, \dots, a_{2k-2}$ are nonzero. But for a_{2k} we would need

$$n \cdot 0 \cdot a_{2k} = -a_{2k-2} \implies 0 \cdot a_{2k} \neq 0,$$

  a **contradiction** (since $a_{2k-2} \neq 0$).
- Thus, when ν is a positive integer, a second Frobenius series solution does **not** exist — a genuinely different construction is needed.



Summary of Solutions for Bessel's Equation

- Bessel's equation of order ν ($\nu \geq 0$):
  $x^2y'' + xy' + (x^2 - \nu^2)y = 0$ has two independent Frobenius series solutions J_ν and $J_{-\nu}$ whenever ν is **not** a non-negative integer:

$$J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n+\nu+1)} \left(\frac{x}{2}\right)^{2n+\nu}, \quad J_{-\nu}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n-\nu+1)} \left(\frac{x}{2}\right)^{2n-\nu}.$$

- So, when ν is not a non-negative integer, the general solution is

$$y(x) = C_1 J_\nu(x) + C_2 J_{-\nu}(x).$$

- When ν **is** a non-negative integer, a second solution independent of J_ν can still be found — called the **Bessel function of the second kind**, Y_ν.
- Then the general solution is

$$y(x) = C_1 J_\nu(x) + C_2 Y_\nu(x).$$



Bessel Functions of the Second Kind

- When $\nu = m$ is a non-negative integer, the Bessel function of the second kind is

$$Y_m(x) = \frac{2}{\pi} J_m(x) \left(\log \frac{x}{2} + \gamma \right)$$
$$+ \frac{x^m}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^{n-1}(h_n + h_{n+m})x^{2n}}{2^{2n+m} n!(n+m)!}$$
$$- \frac{x^{-m}}{\pi} \sum_{n=0}^{m-1} \frac{(m-n-1)!\, x^{2n}}{2^{2n-m} n!}, \qquad x > 0,$$

  where $h_0 = 0$, $h_k = 1 + \frac{1}{2} + \cdots + \frac{1}{k}$, and $\gamma = \lim_{k\to\infty}(h_k - \log k)$ is Euler's constant.

- In particular,

$$Y_0(x) = \frac{2}{\pi} \left[J_0(x) \big(\log(x/2) + \gamma \big) + \sum_{n=1}^{\infty} \frac{(-1)^{n-1} h_n x^{2n}}{2^{2n}(n!)^2} \right].$$

- $Y_0(x)$ behaves like $\log x$ for small x, and $Y_0(x) \to -\infty$ as $x \to 0^+$.



Bessel Functions of the Second Kind: Unified Formula

- For *any* value of ν, the second solution can be unified via a single function $Y_\nu(x)$:

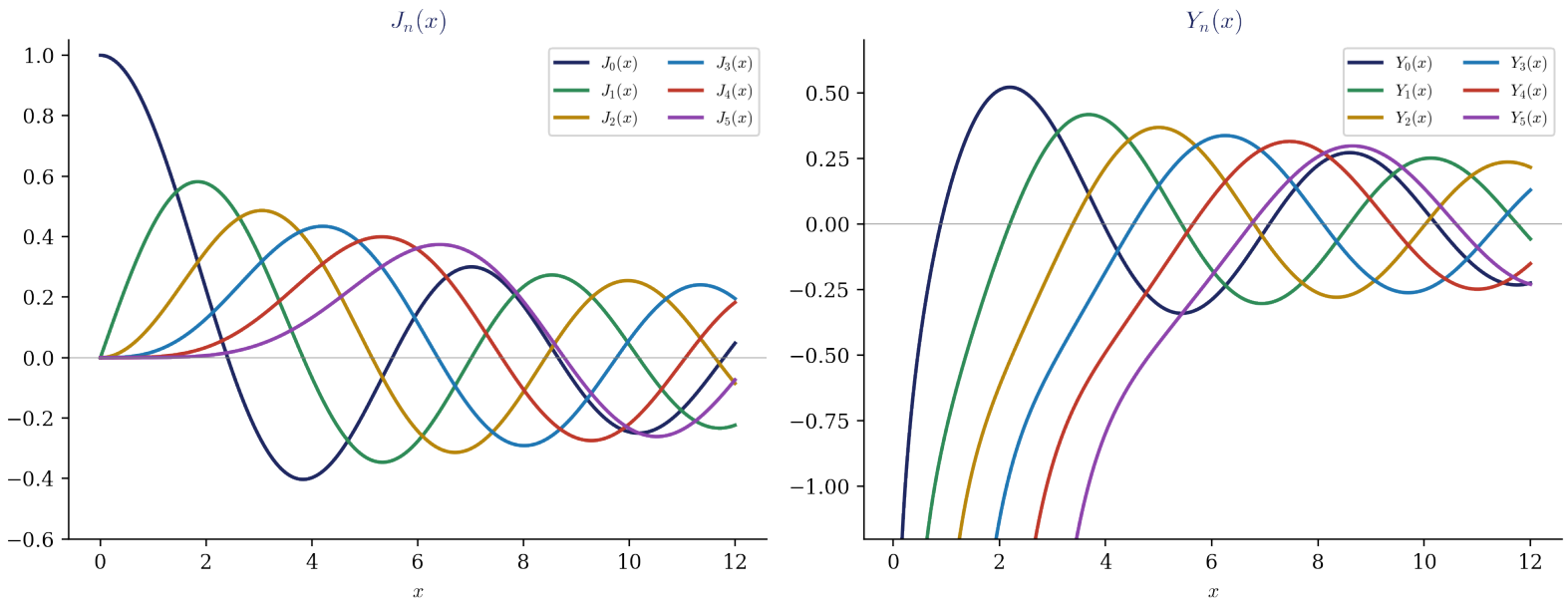
$$\boxed{Y_\nu(x) = \operatorname{cosec}(\nu\pi)\left[J_\nu(x)\cos(\nu\pi) - J_{-\nu}(x)\right]} \qquad (6)$$

  For integer order, define $Y_m(x) := \lim_{\nu\to m} Y_\nu(x)$ (an indeterminate $0/0$ limit).
- When $\nu \notin \mathbb{Z}$, $Y_{-\nu}(x) = aJ_\nu(x) + bJ_{-\nu}(x)$ for some nonzero a, b — so $J_{-\nu}(x)$ is **not** equal to $Y_{-\nu}(x)$ in general.
- $J_\nu(x)$ and $Y_\nu(x)$ are always linearly independent, and $Y_\nu(x)$ always solves Bessel's equation.
- With this function, the general solution of Bessel's equation, valid for **all** $\nu \geq 0$ and $x > 0$, is

$$y(x) = C_1 J_\nu(x) + C_2 Y_\nu(x).$$



Graph of Bessel Functions: $J_n(x)$ and $Y_n(x)$



- $J_n(x)$ is bounded and finite at $x = 0$; $Y_n(x) \to -\infty$ as $x \to 0^+$ for every n.
- Both families oscillate with slowly decreasing amplitude as $x \to \infty$, but are *not* periodic like $\sin x$ and $\cos x$.
- The amplitude of J_n decays like $x^{-1/2}$ — consistent with the asymptotic formula $J_n(x) \sim \sqrt{2/(\pi x)}\cos(x - n\pi/2 - \pi/4)$ for large x.

Linear Dependence of $J_m(x)$ and $J_{-m}(x)$, $m \in \mathbb{N}$

- When $\nu = m$ is a positive integer, using $\Gamma(n+m+1) = (n+m)!$,

$$J_m(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+m)!} \left(\frac{x}{2}\right)^{2n+m}.$$

- Consider $J_{-m}(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n-m+1)} \left(\frac{x}{2}\right)^{2n-m} =$ $\sum_{n=m}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n-m+1)} \left(\frac{x}{2}\right)^{2n-m}$, since each term from $n = 0$ to $n = m-1$ vanishes ($1/\Gamma(k) = 0$ for non-positive integer k).
- Substituting $n - m = k$:

$$J_{-m}(x) = \sum_{k=0}^{\infty} \frac{(-1)^{k+m}}{(m+k)!\,\Gamma(k+1)} \left(\frac{x}{2}\right)^{2(m+k)-m}$$

$$= (-1)^m \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(m+k)!} \left(\frac{x}{2}\right)^{2k+m} = (-1)^m J_m(x).$$

- So J_m and J_{-m} are **linearly dependent** for integer m.



Properties of Bessel Functions: $(x^\nu J_\nu(x))'$

- From (3), $x^\nu J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n+\nu+1)} \left(\frac{x}{2}\right)^{2n+2\nu}$.
- Differentiating with respect to x, and using $\Gamma(n+\nu+1) = (n+\nu)\Gamma(n+\nu)$:

$$(x^\nu J_\nu(x))' = \sum_{n=0}^{\infty} \frac{(-1)^n (n+\nu)}{n!\,\Gamma(n+\nu+1)} \left(\frac{x}{2}\right)^{2n+2\nu-1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma(n+\nu)} \left(\frac{x}{2}\right)^{2n+2\nu-1}.$$

- This is exactly $x^\nu \sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,\Gamma\big(n+(\nu-1)+1\big)} \left(\frac{x}{2}\right)^{2n+\nu-1}$.
- Therefore,

$$\boxed{\big(x^\nu J_\nu(x)\big)' = x^\nu J_{\nu-1}(x)} \tag{7}$$

- It makes integral like $\int x^\nu J_{\nu-1}(x)\,dx = x^\nu J_\nu(x) + C$ immediate — no series manipulation required.



Properties of Bessel Functions: $(x^{-\nu}J_\nu(x))'$

- From (3), $x^{-\nu}J_\nu(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n+\nu}\, n!\, \Gamma(n+\nu+1)} x^{2n}$.
- The $n = 0$ term is constant; differentiating the rest and substituting $k = n - 1$:

$$\left(x^{-\nu}J_\nu(x)\right)' = \sum_{n=1}^{\infty} \frac{(-1)^n}{2^{2n+\nu-1}(n-1)!\, \Gamma(n+\nu+1)} x^{2n-1}$$

$$= -x^{-\nu} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!\, \Gamma\left(k+(\nu+1)+1\right)} \left(\frac{x}{2}\right)^{2k+\nu+1}.$$

- Therefore,

$$\boxed{\left(x^{-\nu}J_\nu(x)\right)' = -x^{-\nu}J_{\nu+1}(x)} \tag{8}$$

- It makes integral like $\int x^{-\nu}J_{\nu+1}(x)\,dx = -x^{-\nu}J_\nu(x) + C$ immediate — no series manipulation required.

